MEDIMCOM IA — autonomous AI agents for safe, lawful and accessible consultations with deaf patients

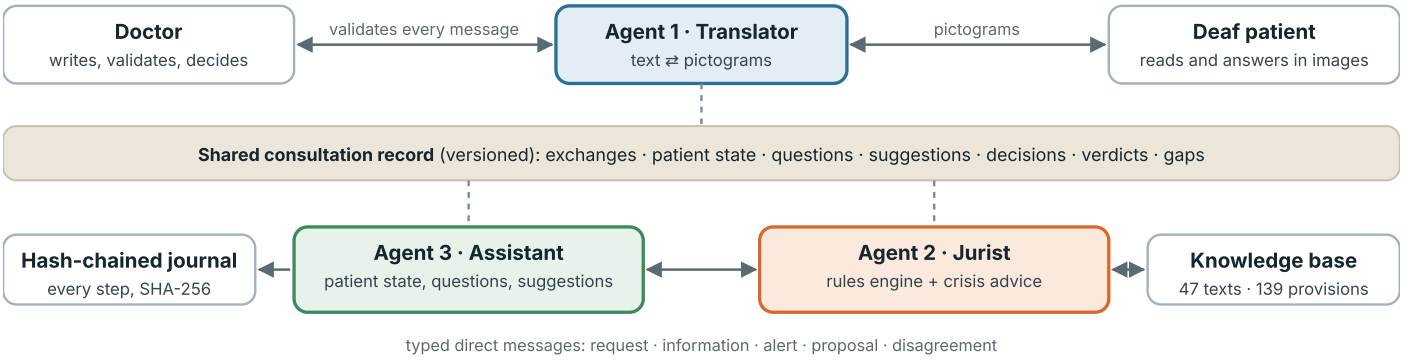
Team **ShifaMiko** · team leader: Baudouin Nkana Bilong (bilongbaudouin2.0@gmail.com · +237 659 296 199); members: Franck Ndonkou, Fotsing Engoulou · Faculty of Sciences, University of Yaoundé I, Cameroon · code: **github.com/baudouin2/MEDIMCOM-IA** · Version 1.2, 9 November 2026 · Prototype in development: official legal texts, synthetic patients only — not a medical device.

MEDIMCOM IA lets a doctor and a deaf patient with limited literacy communicate through pictograms. Autonomous AI agents collaborate around a shared consultation record: a **translator** (text ⇄ pictograms; its pictogram model is in development and plugs in through a fixed interface), an **assistant** (patient state, missing information, suggestions) and a **jurist** that checks every prescription, every message to the patient and every data request against a verified knowledge base of UAE law, and acts as policy and legal advisor in a crisis, where institutional agents (health authority, hospital, crisis cell) coordinate on fragmented information. **The doctor always decides.** Every reasoning step, message and verdict is written to a hash-chained journal.

1 Problem

Deaf patients — especially those who read little — often cannot describe symptoms or understand treatment without an interpreter, and interpreters are scarce, above all in emergencies. Critical facts get lost (pregnancy, allergies, current medicines), consent may not be informed, and instructions are misunderstood. AI can help, but only within strict rules: in the UAE an AI system supports and a human decides (DoH Abu Dhabi AI Policy 2018, Policy statement (AI as supportive technology); DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.7.2.2), health data must stay in the country (UAE Law 2/2019, Art. 13), and every decision must be traceable (DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.7.1.3). In a crisis, information arrives in fragments and rules (notification duties, emergency powers) change what is allowed.

2 Solution: three autonomous agents and one human decision-maker



Agent	Goal and what it decides on its own	Collaboration	Hard limits (enforced in its tools)
1 · Translator	Translates the assistant's questions and every compliant prescription (with a consent question) into pictograms; reports the patient's own pictogram answers (a lone YES/NO to a closed question becomes a fact); flags words with no pictogram instead of inventing.	Proposes each message to the doctor; sends only after the jurist's green light.	Its send tool refuses any message not validated by an authorised, active doctor and judged COMPLIANT, or whose prescription is no longer compliant.
3 · Assistant	Keeps the patient state reliable; spots missing facts; asks questions through the translator; suggests answers or treatments with a source; withdraws obsolete suggestions; raises urgent alerts first.	Answers the jurist's requests; consults the knowledge base before proposing.	Never talks to the patient, never prescribes, no certain diagnosis, cannot confirm a fact nobody reported; the doctor validates or ignores.
2 · Jurist	Checks every prescription, message to the patient and data-sharing request; re-checks when the patient state changes; reminds legal obligations; answers questions in a crisis; records gaps.	Explains verdicts to the doctor; requests missing facts from the assistant; proposes a lawful path.	Cannot change a verdict: verdicts come from a deterministic rules engine; the language model only explains and negotiates.

Why these are agents, not a workflow. Each agent runs its own observe → reason → act loop: it reads its messages and what changed in the shared record since its last turn, and chooses which of its tools to call — or does nothing. The coordinator only releases scenario events and stops when no agent acts any more; it never orders the agents. In the reference runs the jurist re-checks a prescription because the patient state changed, and the assistant withdraws its own suggestion once the answer arrives — nobody tells them to; with a random order of turns the outcomes are identical. Agents run on local open models with tool calling (Ollama, or any OpenAI-compatible server such as llama.cpp or vLLM; one model per agent is possible; target models `qwen3:8b`, `llama3.1:8b`) or on scripted policies; the reference runs below use the scripted policies. The loop tolerates small-model errors (tool calls written as text, mistyped arguments, repeated calls) and degrades gracefully to the scripted policy if the model server

fails. In both modes the guardrails sit in the tools: a tool outside an agent's permissions is refused and logged, the assistant cannot mark a fact as confirmed unless a confirmed exchange reports it, and a coordinator safety net forces the jurist's check on any decision left unchecked.

Interface. A dependency-free local web app connects the agents to two screens: the doctor's (conversation, proposals to validate or reject, prescription form, agents' alerts with sources, crisis board) and the patient's (pictograms only, optional Arabic/English/French labels, large answer palette, a button to combine two pictograms into one image, emergency button); the translator also combines pictograms when clearer (head + pain), never the negation or a dose limit, usable on a tablet; a step-by-step demo mode replays the scenarios.

3 The jurist agent: reasoning separated from decision

Verdicts: **COMPLIANT** · **INSUFFICIENT EVIDENCE** (an unknown fact is never assumed safe) · **REVIEW** (the doctor may proceed with a written justification; his name is logged) · **BLOCKED** (no override in the system). Rules may also attach an **OBLIGATION** (consent, information, notification) or flag a **GAP**. Principle: **only a binding source (law, regulator standard) can block**; a clinical recommendation can at most require review. The 81 rules (`kb/regles_uae.json`) cite provision IDs; a missing fact (consultation mode, data location, written approval) gives INSUFFICIENT EVIDENCE rather than a block; the engine rebuilds each citation (text, article, official link, verification level) from the knowledge base and refuses to start if a rule cites an unknown provision.

Rule family	What the jurist checks	Strongest outcome	Sources (provision IDs in the knowledge base)
Who prescribes	Register of licensed practitioners; author must be the human doctor	**BLOCKED**	UAE Decree-Law 38/2024, Art. 42(2); UAE Decree-Law 4/2016, Art. 4(6); DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.7.2.2
Medicine approval	No marketing approval → blocked; unknown medicine → cannot be checked	**BLOCKED**	UAE Decree-Law 38/2024, Art. 5(1)
History before treatment	Pregnancy (women 12–50), allergies, current treatments must be confirmed	**INSUFFICIENT EVIDENCE**	UAE Decree-Law 4/2016, Art. 4(2)
Written prescription	Dose and usage required	**INSUFFICIENT EVIDENCE**	UAE Decree-Law 4/2016, Art. 4(6)
Pregnancy	Medicine that may cause abortion → blocked; NSAID → review by doctor	**BLOCKED**	UAE Decree-Law 4/2016, Art. 16; FDA NSAID pregnancy safety communication 2020, Recommendation
Telehealth	No controlled/narcotic medicine at a distance (Dubai); identify patient, document consent	**BLOCKED**	DHA Telehealth Standards v4 (2025), Sec. 17.12.7 (also 4.2.2, 17.3.1(e)); UAE Decree-Law 4/2016, Art. 5(5); DHA Telehealth Standards v4 (2025), Sec. 10.1, 10.1.1, 10.4.2
Consent and information	Consent before treatment (emergency exception); explain options and importance, accessibly	**OBLIGATION**	UAE Decree-Law 4/2016, Art. 5(1); UAE Decree-Law 4/2016, Art. 4(5); UAE Law 29/2006, Art. 7; UN CRPD, Art. 21(a)-(b),(e)
AI transparency	Tell the patient AI assists the doctor; patient may opt out	**OBLIGATION**	DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.6.1.1
Messages to the patient	Nothing reaches the patient without validation by an authorised, active doctor	**BLOCKED**	DoH Abu Dhabi AI Policy 2018, Policy statement (AI as supportive technology); DHA AI in Healthcare Policy 2021, Sec. 4.4.1-4.4.2
Health data location	No storage or processing abroad; listed exceptions need written consent, encryption, copy in UAE and a named authorisation	**BLOCKED**	UAE Law 2/2019, Art. 13; UAE Law 2/2019, Art. 24; MOHAP Resolution 51/2021, conditions; DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.4.5.1
Secondary use of data	Health purposes only without written approval; five legal exceptions	**BLOCKED**	UAE Law 2/2019, Art. 16
Communicable diseases	Group A: notify within 24 h; confidentiality of infected persons	**OBLIGATION**	UAE Law 14/2014, Art. 4; UAE Law 14/2014, Art. 29

Crisis mode. Three institutional agents — health authority, reference hospital, crisis cell — each receive only their own fragments of information and never see the patient record. They record measures only on a binding, officially verified provision that confers a power to act (checked by the tool), and every data request they make goes through the jurist. Any of them, or the doctor, can ask the jurist a question. It searches the knowledge base, answers with the applicable provisions — legislation, institutional duties, emergency powers (UAE Law 14/2014, Art. 18), international obligations (WHO International Health Regulations (2005), Art. 6(1); UN CRPD, Art. 11) and global examples — links the relevant gaps, and states that the final decision is human.

4 Knowledge base

Built locally in formats ready for the ITU distributed vector store: `sources.json` (47 instruments with authority, dates, official URL, verification level), `fiches.jsonl` (139 provisions: article, plain summary, short quote, themes), `lacunes.json` (9 gaps), `regles_uae.json` (81 rules), `questions_test.jsonl` (65 test questions) and `index/chunks.jsonl` (148 self-contained chunks with metadata).

Content. UAE host-country sources (15): federal laws on medical liability, ICT in health, personal data, medical products, people of determination and communicable diseases; MOHAP Resolution 51/2021; DoH Abu Dhabi AI policy and Responsible AI Standard; DHA AI policy and telehealth standards; UAE AI Charter; National AI Strategy 2031. Global examples (14): UN CRPD, WHO (AI ethics, LMM guidance, IHR), UNESCO, EU AI Act, Product Liability Directive, MDR, Electronic Communications Code, NHS Accessible Information Standard, MHRA AI Airlock, ISO 9186-1, AU AI Strategy. Plus ITU references and FDA/label clinical references. A second jurisdiction, **Cameroon** (6 texts, own rule file, `--pays CM`), shows the base is transposable.

Verification. 33 sources read on the official publisher's site, 11 through a reliable secondary source, 3 to confirm — the level appears in every citation. Article numbers are cited only when read. Arabic texts prevail. `telecharger_sources.py` downloads the originals and records their SHA-256 on first run; `--verifier` then flags any text changed online.

Retrieval (measured). Dependency-free BM25 with a French→English lexicon, on 65 questions (English and French): recall@1 = 0.80, recall@3 = 0.89, recall@5 = 0.97, MRR = 0.86. The same set will compare multilingual embeddings in the sandbox.

5 Simulated error scenarios (progressive and fragmented)

Scenarios release information fragment by fragment; each agent sees only its messages. **Automated evaluation** (`evaluer_agents.py`): 29 capability and safety expectations plus 6 safety invariants per scenario (checked on the hash-chained journal where possible) — 100 % in reference mode, also with a random order of turns; a **chaotic model** calling random tools never breaks them. **Formal verification**: rules with a written semantics (Kleene logic, verdict lattice, deny by default); 89 (UAE) and 71 (Cameroon) safety and utility properties hold for *all* inputs (exhaustive enumeration + Z3 SMT proofs); a runtime monitor checks 10 invariants over 1,200 adversarial runs; 8 defects found and fixed. Outcomes below come from the reference runs.

Scenario	Fragment of information released	Key outcome	What the agents did on their own
Consultation	Woman, 25, abdominal pain; pregnancy unknown	**COMPLIANT**	Assistant asks “Are you pregnant?” on its own; translator turns it into pictograms; doctor validates; jurist clears; translator sends.
Consultation	Doctor prescribes ibuprofen	**INSUFFICIENT EVIDENCE**	Jurist asks the assistant for the missing fact; the question is already with the patient.
Consultation	Patient taps the YES pictogram	**REVIEW**	Translator records pregnancy = yes; jurist re-checks unasked; assistant proposes a sourced alternative; doctor decides.
Consultation	Doctor prescribes paracetamol (from the AI suggestion)	**COMPLIANT**	Obligations attached (consent, explanation, accessible information); translator prepares the explanation with a consent question; gaps L01, L04.
Consultation	External module sends a message without validation	**BLOCKED**	Blocked; once the doctor validates the translator's version it is cleared and sent (gap L05).
Consultation	Company asks for the list of pregnant patients	**BLOCKED**	UAE Law 2/2019, Art. 16; lawful path stated.
Consultation	Local model outage: send the consultation to a foreign cloud AI?	**BLOCKED**	UAE Law 2/2019, Art. 13, fine under Art. 24; path: model hosted in the UAE or locally (gap L06).
Crisis	Tramadol prescribed by teleconsultation; doctor tries to maintain it	**BLOCKED**	DHA Telehealth Standards v4 (2025), Sec. 17.12.7 (also 4.2.2, 17.3.1(e)); maintaining is refused — a blocking rule has no override.
Crisis	Suspected Group A disease; doctor notifies the authority	**OBLIGATION**	Jurist reminds the 24-hour notification. The authority, on its own, opens an investigation (UAE Law 14/2014, Art. 10), requests surveillance data (cleared), asks the jurist about its powers and then declares an infected area (UAE Law 14/2014, Art. 18).
Crisis	Foreign laboratory requests identified results	**REVIEW**	Listed exception, all conditions met: the doctor authorises it by name with a written reason.
Crisis	Authority asks: is consent given in pictograms valid without an interpreter?	**ADVICE**	Cites UAE disability law, CRPD Art. 21, Dubai telehealth standard and the consent rule; links gap L03 (L04 already recorded).
Crisis	Fragment to the hospital only: a severe case is transferred	**OBLIGATION**	Hospital reserves an isolation bed (legal basis checked), asks the doctor for emergency pictograms, informs the authority (gap L09).

Crisis	Fragment to the crisis cell only: the press wants names	**BLOCKED**	Cell forwards to the authority; its nominative publication is blocked (UAE Law 14/2014, Art. 29); it follows the jurist's path and publishes anonymised figures — COMPLIANT.
Crisis	Fragment to the crisis cell only: wrong pictograms in 3 facilities	**ADVICE**	Cell logs an AI incident, asks the jurist where to report (DHA, DoH, EU AI Act Art. 73 example; gap L08), informs the authority.

6 Policy, strategy and standards gaps

The jurist records a gap when no binding text settles a situation it meets. Each gap lists what UAE texts already say, what is missing, and global policy examples — domain-specific findings, not recommendations to one country. Details: annex *Gap analysis* and `kb/lacunes.json`.

ID	Gap	What UAE texts already say	What is missing	Global examples
L01	**Who answers for an AI suggestion that a clinician validated?** Liability and accountability	UAE Decree-Law 4/2016, Art. 6; DoH Abu Dhabi AI Policy 2018, Policy statement (AI as supportive technology); DHA AI in Healthcare Policy 2021, Sec. 4.2.1-4.2.3	No binding rule allocating liability between clinician, software developer and facility when a validated AI suggestion causes harm, and no rule on the evidential...	EU Product Liability Directive 2024/2853, Arts. 4, 9, 10; EU AI Act, Art. 14
L02	**When is a decision-support or communication assistant a medical device?** Regulatory status of software	UAE Decree-Law 38/2024, Art. 1 (definition of Medical Equipment); DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.3.1.2.2	The decree-law text read does not set criteria separating regulated decision-support software from non-device tools (communication aids, compliance checkers);...	FDA CDS guidance (2026), Four non-device criteria; EU MDR 2017/745, Annex VIII, Rule 11; MHRA AI Airlock, Programme description
L03	**No operational standard for meeting deaf patients' communication needs in care** Accessible health communication	UAE Law 29/2006, Art. 7; DHA Telehealth Standards v4 (2025), Sec. 14.1.5(a), 8.3.5, 14.1.6, 9.6.3; UN CRPD, Art. 21(a)-(b),(e)	No federal operational standard requiring providers to identify, record and flag a patient's communication need, nor rules on which alternative means (pictograms,...	NHS Accessible Information Standard, Standard DCB1605
L04	**Is consent given through pictograms valid informed consent?** Informed consent	UAE Decree-Law 4/2016, Art. 5(1); UAE Decree-Law 4/2016, Art. 8(1)(c); DHA Telehealth Standards v4 (2025), Sec. 10.1, 10.1.1, 10.4.2	No provision recognises consent expressed through augmentative and alternative communication, defines how comprehension is checked, or how such consent is documented.	UN CRPD, Art. 12; UN CRPD, Art. 25; UN CRPD, Art. 21(a)-(b),(e)
L05	**No validation standard for medical pictograms** Standards	No text found in the knowledge base	No UAE standard found on testing the comprehensibility of health pictograms or of AI-generated images used to inform patients.	ISO 9186-1:2014, Scope (Introduction)
L06	**Strict health-data localisation versus cloud AI and international sandboxes** Data governance and innovation	UAE Law 2/2019, Art. 13; MOHAP Resolution 51/2021, list of exceptions; DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.4.5.1	No explicit pathway for AI testing in cross-border or international sandboxes, and no statement on whether synthetic or de-identified data falls outside the...	EU AI Act, Arts. 57-59; MHRA AI Airlock, Programme description
L07	**Oversight rules do not address multi-agent systems** Human oversight and traceability	DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.7.1.3; DHA AI in Healthcare Policy 2021, Sec. 4.4.1-4.4.2; DHA AI in Healthcare Policy 2021, Sec. 4.3.1	No requirement on agent-to-agent communication, limits of agent autonomy, tool permissions, or tamper-evident logs.	EU AI Act, Art. 12; EU AI Act, Art. 14
L08	**No unified channel for AI incidents in health across emirates** Crisis coordination	DHA AI in Healthcare Policy 2021, Sec. 4.6.17, 4.8.1-4.8.2, 4.8.5; DoH Abu Dhabi Responsible AI Standard 2025, Sec. 3.1.7.1.2.3; UAE Law 14/2014, Art. 4	Incident reporting duties are emirate-specific; no federal time limit or shared format for AI incidents found in the knowledge base.	EU AI Act, Art. 73; WHO International Health Regulations (2005), Art. 6(1)
L09	**Emergency protocols do not guarantee equivalent access for deaf people** Accessible emergency communication	DHA Telehealth Standards v4 (2025), Sec. 12.4-12.4.3, 14.1.4(c); UAE Law 29/2006, Art. 7; UN CRPD, Art. 11	No requirement found that health emergency protocols provide equivalent non-voice access for deaf patients.	EU Electronic Communications Code 2018/1972, Art. 109(5)

7 Auditability

- **Hash-chained journal.** Each line holds time, agent, type (event, reasoning, action, message, verdict, gap), a frozen copy of the content, the previous hash and its own SHA-256 hash. Any later change to a line or to the order is detected (`verifier_journal.py --falsifier 10` shows it live). The first line records the configuration (models, temperature, order of

turns, hashes of the rules and knowledge base) for reproducibility; `exporter_journal.py` turns any journal into a filterable HTML timeline.

- **Reasoning traces.** Each agent's reasoning is logged before its actions; every tool call is logged with its arguments and result, including refused calls.
- **Source attribution.** Every verdict and obligation lists provision IDs with the official link and verification level; every assistant suggestion carries its source ID.
- **Named human decisions.** Maintaining a REVIEW decision requires an authorised, active doctor and a written justification, both logged; a BLOCKED decision cannot be maintained. Journals are saved as JSON files the facility can archive for the 25-year retention required by UAE Law 2/2019, Art. 20(1).

8 Use case on the ITU-T Y.3172 pipeline, mapped to the ITU AI Readiness Framework 2.0

Each pipeline node lists the concrete data or function, the host-country documents that govern it, and the readiness factor · dimension (Report 2.0, January 2026) with the metric this prototype reports. All six factors are covered (data, research, deployment support, standards, open source and code, sandbox).

Node	MEDIMCOM IA (data, how collected, how preprocessed)	Governing documents (knowledge base)	Readiness factor · dimension → metric
SRC	Doctor's typed or dictated text (FR/EN); patient's pictogram IDs (JSON list, e.g. VENTRE, DOULEUR, CHIFFRE:7); record form (age, weight, pregnancy, allergies, treatments); photos/PDFs of prior prescriptions sent by the patient; 139 provisions with URL and verification level. Fictitious patients only.	ICT in Health Law 2/2019 (data stay in the UAE); PDPL 45/2021; People of Determination Law 29/2006	Data · D1 data/model marketplace → 47 open sources with links; D11 AI for inclusion → text-free input
C	Local server on the clinic computer: JSON events (tap, session ID, time); patient documents ≤ 10, ≤ 12 MB, de-duplicated by SHA-256; official texts downloaded with frozen SHA-256	DHA Telehealth Standards v4 (accessible channel, emergency)	Deployment support · D6 human interface → image-only mode, fr/en/ar labels
PP	Cleaning: typo/dictation correction, dose normalisation; document facts stay *unconfirmed* until the doctor confirms; de-identification: names and IDs stripped from messages to institutions; fictitious data only with an online model; 148 legal chunks	PDPL 45/2021; MOHAP Res. 51/2021 (transfers abroad)	Data · D10 AI and policies → data sovereignty: local processing, no transfer abroad
M	Language model for the assistant: local open-source (Ollama) or any OpenAI-compatible API (one key in `CLE_API.txt`); BM25 retrieval (recall@3 = 0.89); 81 deterministic rules	DoH Responsible AI Standard 2025; DHA AI Policy 2021	Research; Open source · D8 collaboration with AI → capability score vs scripted baseline
P	Jurist verdicts; doctor validates every message; named override; institutions need a binding basis in their own mandate	Medical Liability Law 4/2016; UAE AI Charter (human oversight)	Standards · D10 AI and policies → policy tested in 3 sandbox scenarios; 9 gaps
D	Typed agent messages; pictograms sent only after validation and the jurist's check	DHA Telehealth Standards (consent, explanation)	D7 strategy alignment → institutions coordinate through mandates
SINK	Doctor and patient screens; hash-chained journal; crisis board	Communicable Diseases Law 14/2014 (24-hour notification)	D11 AI for inclusion → comprehension rate of pictograms (ISO 9186-1 test, to run)
MLFO	Coordinator, simulated mode, evaluation bench (6 safety invariants), 87 automated tests	ITU-T Y.3172	Sandbox · D13 digital infrastructure → runs offline on one laptop

Reference links of every material proposed to the knowledge base: `docs/soumission/KB_reference_links.md`.

9 Limits and next steps

- Reference results come from scripted policies; the model modes run with local Ollama models and online APIs (Groq, open-source models) and are tested with mock and chaotic models, but no model benchmark is claimed. The jurist never depends on the model.
- Only 7 documented medicines are checked (others → insufficient evidence, no override); clinical rules come from public sources and must be validated by a physician against the UAE formulary. Practitioners are fictitious.
- 11 sources are known through secondary sources and 3 remain to confirm (Cameroon); Arabic originals are not yet ingested.
- Agent 1 uses placeholder pictograms (engine interface: `docs/technique/CONTRAT_AGENT1.md`); prototype security only (doctor token, no HTTPS). **No user test has been run yet**: the protocol, printable booklet and analysis script (comprehension test inspired by ISO 9186-1, at least 20 deaf participants with an interpreter) are ready; only measured results will be claimed.

10 Key references

UAE Federal Decree-Law 4/2016 (Medical Liability) · UAE Federal Law 2/2019 (ICT in Health) · UAE Federal Decree-Law 45/2021 (PDPL) · UAE Federal Law 29/2006 (People of Determination) · UAE Federal Law 14/2014 (Communicable Diseases) · DoH Abu Dhabi Responsible AI Standard 2025 · DHA Telehealth Standards v4 (2025) · ITU-T Y.3172 · ITU AI Ready Report 2.0 (2026). Official links of all 47 sources: `docs/soumission/KB_reference_links.md`.